

Final assessment: Dataset proposal

Design, Implementation, Evaluation, and
Interpretation of Regression and
Classification Models.

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Date: 24 Jan, 2026

Classification Task

Dataset Description

The dataset used for the classification task is **Nutrition, Physical Activity, and Obesity – Youth Risk Behavior Surveillance System (YRBSS)**.

This dataset is collected through surveys conducted among students in grades 9–12 in the United States. It contains information about students' nutrition habits, physical activity levels, and obesity-related behaviors. The dataset includes demographic features such as sex, grade, race/ethnicity, and location, along with survey-based behavioral indicators.

Alignment with UNSDG Goals

This dataset aligns mainly with:

- **SDG 3 – Good Health and Well-Being:**
The dataset focuses on adolescent health, nutrition behavior, physical activity, and obesity prevention.
- **SDG 2 – Zero Hunger:**
Nutrition-related questions such as fruit and vegetable consumption directly relate to healthy diet access.
- **SDG 10 – Reduced Inequalities:**
The dataset includes stratification by race, sex, grade, and location, helping to analyze health inequalities among youth.

Why I Chose This Dataset

I chose this dataset because it is well-structured and highly suitable for **classification tasks**. Most of the important features are categorical, such as grade, sex, and race/ethnicity, which makes it ideal for predicting classes or categories. The dataset is collected from a trusted source and represents real-world public health data, making it meaningful for applied machine learning.

How I Will Use This Dataset for Training

I will use this dataset to build a classification model that predicts **youth risk behavior categories**, such as unhealthy nutrition behavior or low physical activity levels.

The dataset will be preprocessed by handling missing values, encoding categorical variables, and splitting the data into training and testing sets. Classification algorithms such as Logistic Regression or Decision Trees will be used to train the model and evaluate its performance.

Regression Task

Dataset Description

The dataset used for the regression task is the **Existing Buildings Energy & Water Efficiency (EBEWE) Program** dataset.

This dataset contains information about building-level energy use, water consumption, carbon dioxide emissions, and compliance status. It includes numeric features such as building floor area, year built, occupancy, energy use intensity, and emissions data. The dataset is collected as part of a city-level sustainability and energy benchmarking program.

Alignment with UNSDG Goals

This dataset aligns with the following UNSDGs:

- **SDG 11 – Sustainable Cities and Communities:**
The dataset focuses on improving energy efficiency and sustainability in urban buildings.
- **SDG 13 – Climate Action:**
Carbon dioxide emissions and energy efficiency indicators directly support climate action analysis.
- **SDG 7 – Affordable and Clean Energy:**
Energy usage and efficiency metrics relate to responsible and efficient energy consumption.

Why I Chose This Dataset

I chose this dataset because it contains **continuous numerical variables**, which are suitable for regression analysis. The dataset represents real-world environmental and sustainability data and allows meaningful prediction of energy use and emissions. It also reflects real policy-driven data, which makes the analysis more practical and realistic.

How I Will Use This Dataset for Training

I will use this dataset to train a regression model that predicts **carbon dioxide emissions or energy use intensity** based on building characteristics.

The data will be cleaned by converting string-based numeric values into numerical form and handling missing values. The dataset will then be split into training and testing sets. Regression models such as Linear Regression or Random Forest Regression will be applied to predict the target variable.

Dependent and Independent Variables

Dependent Variable (Target)

- Carbon Dioxide Emissions (Metric Ton CO₂e)
(or alternatively Site Energy Use Intensity)

Independent Variables (Features)

- Gross Building Floor Area
- Year Built
- Occupancy
- Number of Buildings
- Property Type
- Program Year
- Postal Code